Evolutionary Game Theory: Individual Project
The Evolution of Cooperation and Punishment through Commitment

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1 Introduction

The willingness of experimental subjects to engage in costly punishment to sanction free riders in the Prisoner's Dilemma is a robust and widely replicated finding in behavioral economics (Chaudhuri, 2011; Fehr & Gächter, 2000). These behaviors are typically interpreted through the lens of reciprocity theory—according to which cooperation emerges because individuals are inclined to reward kind behavior and retaliate against unkind behavior.

However, reciprocity alone does not seem sufficient to explain certain observed patterns of human prosociality. For instance, individuals often choose to punish even when such actions have no strategic consequences, suggesting the presence of an intrinsic motivation or a "taste for punishment" (Dreber et al., 2008; Fudenberg & Pathak, 2010).

An alternative explanation stems from the theory of commitment (Frank, 1988). Within this framework, punishment is not driven by immediate payoffs but by its signaling value. A commitment to punish—even at a personal cost—can credibly signal to others that defection will be met with retaliation, thereby sustaining cooperation over time.

This project investigates the evolutionary stability of commitment-based strategies in a finite population using a Moran process. The underlying game is a two-stage repeated Prisoner's Dilemma with opportunities for costly punishment. By comparing the performance of commitment-based strategies against more conventional strategies, the analysis aims to shed light on the conditions under which commitment can evolve and stabilize within a population.

Game Description

The game unfolds in two stages. In the first stage, players choose whether to cooperate (C) or defect (D). Payoffs follow the classic Prisoner's Dilemma structure:

	C	D
C	2	0
D	3	1

In the second stage, players may choose to punish the opponent (P) at a personal cost of 1, reducing the opponent's payoff by 2. Alternatively, they can choose not to punish (N). The punishment payoff matrix is:

	P	N
P	-3	-1
N	-2	0

Strategies

I restrict attention to five distinct strategies, each characterized by different rules for cooperation and punishment. These strategies determine behavior in both stages of the game: the initial Prisoner's Dilemma and the subsequent punishment game.

- **DNN** (Defect–Never punish): Always defects in the Prisoner's Dilemma and never punishes the opponent, regardless of their behavior.
- **CNN** (Cooperate–Never punish): Always cooperates in the Prisoner's Dilemma but does not punish, even if the opponent defects.

- **CPN** (Cooperate–Punish if necessary): Cooperates in the first stage and punishes the opponent in the second stage if and only if the opponent defected.
- **FNN** (Conditional cooperation–Never punish): Cooperates only if the opponent is punish defection; otherwise, defects. Never punishes the opponent, even when they defect.
- **FPN** (Conditional cooperation–Punish if necessary): Behaves like FNN in the first stage, cooperating only if the opponent is a punisher. In addition, it punishes opponents who defect.

The last two strategies—**FNN** and **FPN**— are possible only if we allow commitment as a mechanism for the evolution of cooperation. More precisely, I assume that agents are able to send a signal (which I assume to be perfectly credible) to communicate their punishing behavior. Moreover, strategies FNN and FPN are able to condition their cooperative behavior on this signal.

The following matrix summarizes the aggregate payoffs for each strategy pair. These payoffs combine both stages of interaction and represent the discounted, normalized returns from the infinitely repeated two-stage game:

	DNN	CNN	CPN	FNN	FPN
DNN	1	3	1	1	−1
CNN	0	2	2	0	−2
CPN	−1	2	2	2	2
FNN	1	3	2	1	0
FPN	0	3	2	3	2

2 Analysis in the Limit of Low Mutation Rates

2.1 Fixation Probabilities

In the Moran process, the fixation probability of a mutant of type A invading a population of type B is given by (Nowak, 2006):

$$\rho_{AB} = \frac{1}{1 + \sum_{k=1}^{N-1} \prod_{i=1}^{k} \frac{g_i}{f_i}}$$

where fitness values are computed as:

$$f_i = 1 - w + w \cdot \frac{a(i-1) + b(N-i)}{N-1}, \qquad g_i = 1 - w + w \cdot \frac{ci + d(N-i-1)}{N-1}$$

Here, w denotes the intensity of selection, and a, b, c, d are payoffs from the 2*2 pairwise matrix obtained from the matrix above.

Using simple numerical methods, I compute the pairwise fixation probabilities for all strategy combinations. I set initially set the parameters $N = 500, w = 0.9$. The results are displayed in the following table, where the row represents the mutant and the column the resident strategy:

	DNN	CNN	CPN	FNN	FPN
DNN	0.000000	0.323588	0.000000	0.002000	0.000000
CNN	0.000000	0.000000	0.002000	0.000000	0.000000
CPN	0.000000	0.002000	0.000000	0.471667	0.002000
FNN	0.002000	0.323588	0.000000	0.000000	0.000000
FPN	0.000000	0.328317	0.002000	0.643731	0.000000

2.2 Transition Matrix under Rare Mutations

Using the fixation probabilities, I construct the transition matrix of a Markov chain over the five strategies in the limit of rare mutations (Fudenberg and Imhof, 2006). Each off-diagonal element T_{ji} gives the probability of moving from strategy j to i and is computed as:

$$T_{ji} = \frac{\mu}{n-1} \cdot \rho_{ij}, \qquad \text{for } i \neq j$$

Here, μ is the mutation rate (set at 0.1), n is the number of strategies, and ρ_{ij} is the fixation probability of strategy i invading strategy j. The diagonal terms ensure each row sums to 1:

$$T_{jj} = 1 - \sum_{i \neq j} T_{ji}$$

The resulting matrix represents the probability of transitioning from a population of individuals playing strategy j to another population playing strategy i under the limit of low mutations. Each row of the matrix below corresponds to the current strategy, and each column to the strategy that would take over after a successful mutation and fixation:

	DNN	CNN	CPN	FNN	FPN
DNN	0.999950	0.000000	0.000000	0.000050	0.000000
CNN	0.008090	0.975563	0.000050	0.008090	0.008208
CPN	0.000000	0.000050	0.999900	0.000000	0.000050
FNN	0.000050	0.000000	0.011792	0.972065	0.016093
FPN	0.000000	0.000000	0.000050	0.000000	0.999950

2.3 Stationary Distribution

The transition matrix computed above it's ergodic. Taking the normalized eigenvector associated with eigenvalue 1 gives the stationary distribution of the stochastic process. This is reported in the table below for the process characterized by the parameters: $N = 500, w = 0.9, \mu = 0.1$.

	Stationary Dist.
DNN	0.108804
CNN	0.000670
CPN	0.327499
FNN	0.000389
FPN	0.562638

This result reveals that in the long run populations frequently end up dominated by FPN more than any of the other strategies considered. This suggests that individuals who both condition their behavior on others' willingness to punish and actively punish defectors tend to outcompete other strategies. The combination of contingent cooperation and punishment proves evolutionarily stable. More broadly, the possibility of being committed to punishment may help explain the evolution of cooperative attitudes.

3 Beyond the Limit of Low Mutation

Under construction ...

References

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